FLOWOID TECHNOLOGIES

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Ayanshi Imitation

Business Management System

Project Specification & Implementation Plan

Version 1.0 | Prepared by: Flowoid Technologies

1. Project Overview

1.1 About the Client

Client Name: Ayanshi Imitation

Industry: Imitation Jewellery Manufacturing & Wholesale Distribution

Business Type: B2B – Wholesale to registered dealers

Ayanshi Imitation is an imitation jewellery manufacturing business that sources raw materials from suppliers, processes and manufactures jewellery through a distributed artisan/worker model, and sells finished products wholesale to dealers. The business is entirely B2B — no direct retail sales to end consumers.

1.2 About Flowoid Technologies

Flowoid Technologies is the software development partner responsible for designing, developing, and delivering this management system. This client engagement is the first production deployment. The long-term product vision is a scalable multi-tenant SaaS platform configurable for various manufacturing and wholesale businesses across India.

1.3 Technology Stack

Layer	Technology
Database	PostgreSQL
Backend / API	Elysia.js (Bun runtime)
Server	Oracle VPS (Linux)
Authentication	JWT with Role-Based Access Control (RBAC)

2. End-to-End Business Flow

The complete operational flow of Ayanshi Imitation is as follows:

Step	Stage	Description
1	**Raw Material Purchase**	Owner purchases raw material from suppliers. Material is recorded with quantity and cost.
2	**Lot Distribution**	Raw material is split into smaller lots. Each lot is issued to a specific worker along with required supplementary materials (stones, diamonds, fittings, coatings, etc.).
3	**Manufacturing**	Workers manufacture jewellery pieces per their assigned design. An efficient worker may hold multiple active design assignments simultaneously.
4	**Goods Collection**	Workers return finished pieces. Quantity received is recorded per assignment. Worker payment is calculated: pieces delivered x rate per piece (rate is determined by diamond count of the design).
5	**Packaging**	Owner packages finished pieces into dozens. This converts unpackaged stock into sellable packaged inventory.
6	**Order & Dispatch**	A dealer places an order. A parcel is prepared from packaged stock and dispatched.
7	**Payment Collection**	Payment collected as cash or credit (udhaar). Credit orders track due date and outstanding balance.

3. User Roles & Permissions

The system implements Role-Based Access Control (RBAC). Three roles are defined at launch:

3.1 Owner

The business owner has unrestricted access to the entire system. This is the super-admin role.

- **Full access to all modules: Dashboard, Analytics, Raw Material, Workers, Inventory, Orders, Parties, Finance**
- **Create, edit, and delete all records**
- **View all financial reports and outstanding balances**
- **Manage users — create and deactivate staff accounts**
- **Configure system settings — credit limits, piece rate tiers, stock alert thresholds**
- **Override credit limit warnings when approving dealer orders**

3.2 Manager / Staff

Operational staff who handle day-to-day data entry and processing. Limited to operational modules only — no access to financial configuration or user management.

- Record raw material issuance to workers
- Record finished goods received from workers
- Create and process sales orders
- Prepare and dispatch parcels
- Record cash payments against orders
- Read-only access to inventory and stock levels
- Cannot access worker payment settlement or supplier financial records
- Cannot manage users or system settings
- Cannot delete records — only create and update

3.3 Viewer / Auditor

Read-only access for the owner's accountant, auditor, or silent partner who requires visibility without modifying data.

- Read-only access to all modules
- Can view and export all reports (PDF, Excel)
- Cannot create, edit, or delete any record
- Cannot see system settings or user management

4. System Modules & Features

4.1 Dashboard & Analytics

The central overview screen. Visible immediately after login for Owner and Manager roles.

- Today's sales summary — orders dispatched, revenue collected
- Low stock alerts — designs with packaged stock below threshold
- Outstanding payments — top overdue dealers with amounts
- Worker activity — active assignments, pending return quantities
- Raw material stock level overview
- Monthly sales trend chart — units sold and revenue
- Worker productivity chart — pieces collected per worker this month
- Top-selling designs by quantity this period

4.2 Raw Material Management

Tracks all incoming raw material purchased from suppliers.

- Record raw material purchase: supplier, material type, quantity (kg/units), unit cost, date
- Current raw material stock — total purchased minus total issued to workers
- Raw material categories (metal base, gold plating, silver plating, rhodium, etc.)
- Supplier ledger — total purchased, amount paid, outstanding balance to supplier
- Low stock alert per material type

4.3 Supplementary Materials

Supplementary materials are items issued to workers alongside raw material lots — stones, diamonds, fittings, threads, coatings, and other accessories.

- Maintain a catalogue of supplementary material types
- Track stock quantity for each supplementary item
- Record issuance per worker assignment
- View total consumption per design and per period

4.4 Product Design Catalogue

The master list of jewellery designs. This is the core reference used by the worker, inventory, and order modules.

- Add and manage designs with a unique design code / SKU
- Design attributes: category (necklace, earring, bracelet, ring, maang tikka, etc.), material, finish type
- **Diamond count per design — this is the key driver of worker piece rate**

- Worker piece rate — set manually or auto-calculated from diamond count tiers configured by owner
- Sale price per dozen — default wholesale price
- Attach reference image for each design (optional)
- Mark designs as active or discontinued

4.5 Worker Management

One of the most critical and complex modules. Manages the full lifecycle of manufacturing through external artisan workers.

Worker Profiles

- Register workers: name, phone, address, ID proof type and number
- View full worker history — all assignments, pieces collected, total earnings, pending payments
- Mark workers as active or inactive

Assignment & Lot Issuance

- Assign one or more designs to a worker
- Issue raw material lot (quantity) and supplementary materials per assignment
- Record issue date and expected return date
- Assignment status lifecycle: Issued → In Progress → Partially Returned → Completed

Finished Goods Collection

- Record pieces received from a worker per assignment
- Support partial returns — worker may deliver in multiple batches
- Auto-calculate earnings: pieces received x piece rate for that design
- Track remaining quantity still pending from the worker
- Option to mark returned pieces as rejected (quality issue) — excluded from payment and stock

Worker Payment Settlement

- View total accumulated earnings per worker
- Record payments made to worker (cash)
- Outstanding = total earnings – payments made
- Advance payment support — advance deducted from future earnings
- Full payment history per worker

4.6 Inventory Management

Two-stage inventory: unpackaged finished pieces and packaged dozens ready for sale.

Unpackaged Stock

- Auto-updated when finished pieces are collected from workers
- Shows count of loose finished pieces per design, not yet packed

Packaged Stock

- Owner converts unpackaged pieces into packaged dozens
- Record packaging batch: design, number of dozens, date packed
- Packaged stock is the sellable unit — deducted when orders are dispatched
- Configurable low-stock alert threshold per design

4.7 Party Management (Dealers & Suppliers)

Unified management for all external parties — both dealers (buyers) and suppliers.

- Register dealers: name, city, phone, GST number, credit limit, credit period (days)
- Register suppliers with similar profile details
- Opening balance support for onboarding existing parties with prior history
- Full chronological ledger per party
- Outstanding balance always visible on party profile
- Credit limit enforcement — system warns if a new order exceeds dealer credit limit
- Dealer-specific pricing — override default catalogue price per design if negotiated differently
- Overdue invoice alerts for credit dealers

4.8 Orders & Dispatch

Manages the complete wholesale order lifecycle from creation to dispatch.

- Create sales order linked to a dealer
- Add line items: design, quantity in dozens, price per dozen
- Stock validation before order confirmation — blocks if packaged stock is insufficient
- Generate printable order summary and invoice
- Order status: Draft → Confirmed → Packed → Dispatched
- Record dispatch details (transport mode, tracking reference if applicable)
- Packaged stock auto-deducted on dispatch
- Partial dispatch support — invoice only for dispatched portion

4.9 Payment & Ledger Management

Handles all money movement in and out of the business.

- Record dealer payment against invoice — full or partial
- Payment modes: cash, bank transfer, UPI
- Credit orders: due date auto-set from dealer credit period
- Outstanding aging report — 0-30, 31-60, 61-90, 90+ days overdue
- Supplier payment recording — track outstanding to suppliers

- Worker payment recording — integrated with worker settlement module
- Daily cash flow summary — total inflow vs outflow
- Advance payment handling for dealers and workers

4.10 Reports & Analytics

- Sales report — filterable by date range, dealer, design
- Stock report — current packaged stock, unpackaged stock, raw material levels
- Worker report — pieces collected, earnings, outstanding per worker for any period
- Outstanding report — dealer-wise overdue invoices with aging
- Purchase report — raw material and supplementary purchases
- Profit overview — revenue vs material cost vs worker payment cost
- Export all reports to PDF and Excel

5. Edge Cases & Business Rules

5.1 Worker Module

Scenario	System Behaviour
Worker returns fewer pieces than issued	Partial return recorded. Assignment stays open. Remaining quantity tracked as pending. Worker can make further deliveries.
Worker holds multiple active assignments	Each design creates an independent assignment record. Tracked and paid separately.
Rejected / defective pieces returned	Owner marks pieces as rejected. Excluded from payment calculation and not added to sellable stock. Recorded in a separate rejection log.
Worker paid in advance	Advance recorded. Deducted automatically from next payment settlement.
Worker becomes inactive mid-assignment	Assignment flagged as open with inactive worker. Owner manually closes after final collection or write-off.
Design discontinued mid-assignment	Existing assignments continue until completion. Design marked discontinued only blocks new assignments.

5.2 Inventory & Orders

Scenario	System Behaviour
Order quantity exceeds packaged stock	System blocks confirmation and displays stock shortage per design. Owner can split the order.
Order partially dispatched	Partial dispatch recorded. Invoice for dispatched items. Remainder held as pending.
Order cancelled after dispatch	Cancellation with reason recorded. Stock reversed. Payment adjusted if already collected.
Dealer has custom pricing	Dealer-specific price overrides catalogue price per design on new orders for that dealer.
Same design ordered at different prices by same dealer	Each order line records its own price at time of order. Historical invoices are not affected.

5.3 Payments & Finance

Scenario	System Behaviour

Dealer exceeds credit limit	System shows a warning. Owner can override and approve the order with a logged reason.
Payment before invoice (advance)	Recorded as advance against dealer account. Auto-applied to the next invoice raised.
Partial payment on credit invoice	Partial payment recorded. Remaining balance stays outstanding. Due date unchanged.
Overpayment by dealer	Excess credited to dealer account. Auto-applied to the next invoice.
Supplier payment dispute	Payment can be recorded with a note/remark field for dispute tracking.

6. Future Scope – SaaS Roadmap

While this system is built exclusively for Ayanshi Imitation as a pilot client, Flowoid Technologies plans to evolve this into a multi-tenant SaaS platform. The following capabilities are planned for future releases:

6.1 Multi-Tenancy

- Each business operates in a fully isolated tenant environment
- Subscription-based plans: Basic, Standard, Premium
- Dynamic module enablement per plan — clients only see what their plan includes
- Per-tenant branding — logo and business name
- Current schema is designed with tenant_id on all key tables, making migration seamless

6.2 Premium Features (Planned)

- WhatsApp automated messaging — order confirmations, payment reminders, dispatch notifications
- Bulk WhatsApp broadcast to dealers for new stock arrivals
- Mobile-optimised PWA for owner and staff
- Digital shareable catalogue link for dealers to browse available stock
- QR / barcode generation per design for fast order scanning
- GST-compliant invoice generation
- Tally export for accounting integration
- Advanced analytics — seasonal demand forecasting, best-performing designs
- Multi-branch support for businesses operating from multiple locations

7. Implementation Plan

Phase	Module	Deliverables	Status
1	**Foundation**	Database schema design, PostgreSQL setup on Oracle VPS, Elysia.js project scaffold, JWT Auth + RBAC middleware	Planned
2	**Manufacturing Core**	Raw Material, Supplementary Materials, Design Catalogue, Worker Management (issuance, collection, payment)	Planned
3	**Sales Flow**	Inventory (unpackaged + packaged), Party Management, Orders & Dispatch	Planned
4	**Finance & Reports**	Payment & Ledger, Outstanding tracking, All Reports, PDF/Excel export	Planned
5	**Hardening & UAT**	Edge case testing, performance tuning, security review, client user acceptance testing, go-live	Planned